

AIM:- **Exchange the contents of these two blocks**

- A) A block of data is stored in memory location from 7100H to 7109H. Another block of data having the same length is stored in memory location starting from 7200H. Write a program to Exchange the contents of these two blocks.
- B) Enter the program on microprocessor kit.
- C) Execute the program. Write the contents of both data blocks before and after execution as well as the contents of the registers used in the program after execution and the bit contents of five flags individually. Verify the results.

ALGORITHM:-

1. Start.
2. Initialize HL register pair by 7100H.
3. Initialize DE register pair by 7200H.
4. Initialize Register C by 0AH.
5. Copy the contents of memory location pointed by HL pair to B register.
6. Copy the contents of memory location painted by DE pair to accumulator.
7. Copy the contents of accumulator to memory location pointed by HL pair.
8. Copy the contents of B register to accumulator.
9. Store the contents of accumulator into memory location pointed by DE pair
10. Increment HL pair by 1.
11. Increment DE pair by 1
12. Decrement C register by 1.
13. If C $\neq$ 0 then go to step 5.
14. Stop.

8085 PROGRAM

Memory Location	Label	Hex Code	Mnemonics	Comments
7000H		21	LXIH, 7100H	Copy 7100H into HL register pair
7001H		00		
7002H		71		
7003H		11	LXI D, 7200H	Copy 7200H into DE register pair
7004H		00		
7005H		72		
7006H		0E	MVI C, 0AH	Copy 0AH into C register
7007H		0A		
7008H	Back	46	MOV B,M	Copy the contents of memory Location pointed by HL pair to B register.
7009H		1A	LDAX D	Copy the contents of memory location pointed by DE pair to accumulator.
700AH		77	MOV M,A	Copy the content of accumulator to memory location pointed by HL pair
700BH		78	MOV A,B	Copy the contents of B register to accumulator
700CH		12	STAX D	Store the contents of accumulator to Memory location pointed by DE pair
700DH		23	INX H	Increment HL pair by 1
700EH		13	INX D	Increment DE pair by 1
700FH		0D	DCR C	Decrement C register by 1
7010H		C2	JNZ Back	Jump on Back If Register C ≠ 0
7011H		08		
7012H		C0		
7013H		CF	RST 1	Stop

INPUT OUTPUT:-

Memory Location	Before Execution	After Execution
7100H	00H	11H
7101H	01H	12H
7102H	02H	13H
7103H	03H	14H
7104H	04H	15H
7105H	05H	16H
7106H	06H	17H
7107H	07H	18H
7108H	08H	19H
7109H	09H	20H

Memory Location	Before Execution	After Execution
7200H	11H	00H
7201H	12H	01H
7202H	13H	02H
7203H	14H	03H
7204H	15H	04H
7205H	16H	05H
7206H	17H	06H
7207H	18H	07H
7208H	19H	08H
7209H	20H	09H

REGISTER VALUES:-

Register	Contents	Register	Contents
A	09H	F	54H
B	09H	C	00H
D	72H	E	71H
H	71H	L	0AH

FLAG REGISTER:- 54H

S	Z	X	AC	X	P	X	CY
0	1	0	1	0	1	0	0

CONCLUSION:-

Thus we have executed the program for the exchange of two memory block.